

Supplementary Information

Target-conditioned refinement of future-blind longitudinal glioma change maps replicates across independent cohorts

[AUTHOR_LIST_REQUIRED]

Supplementary Methods

This supplement expands reproducibility, cohort-flow, ablation, robustness, secondary metric and reporting details. All values are deterministic summaries of frozen releases; no model or correction method was rerun.

Evidence provenance

Development authority was PCC_LEAKAGE_FREE_RERUN_2026 Kaggle version 8 plus the 2026 internal completion and validity patch. Internal confirmation used the Stage B V2 authority, whose V1 scientific CSVs were proven unchanged. RHUH used the pre-outcome protocol lock, physically isolated Stage A P0 freeze and Stage B release. SHA-256 values are listed in Supplementary Table S1 and the authority registry.

Cohort amendments

The 113-patient cohort amendment excluded PatientID_0113 and PatientID_0132 before target construction or performance access because current T1c/current masks and future T1c were identical while future masks differed. Both were excluded rather than selecting by labels or results. All original P0 evidence was retained. RHUH-0008 was excluded before external P0 generation because lossless orientation operations did not establish a shared physical grid. No registration, interpolation or case-specific header repair was allowed.

Development controls

Removing error guidance, removing outside suppression, no smoothing, global discrepancy, shuffled targets and imperfect guidance were evaluated without changing the frozen P0. Shuffled donors were patient-level. Imperfect-guidance repeats were aggregated within patient. No-smoothing was not the original canonical method and its development observation was explicitly treated as post-hoc.

Spatial analyses

Layer 1 v1 was selected on provenance, not performance magnitude; v1.1 is sensitivity evidence. Layer 3 used two-sided tests and prespecified Holm families. Absolute core PRI and peritumour-control localization survived their respective corrections; several boundary and relative effects were nominal or unsupported. The manuscript therefore treats these analyses as exploratory spatial reliance/localization, not causal biology.

Supplementary Results

Supplementary Table S1. Authority packages

See `PCC\_MANUSCRIPT\_AUTHORITATIVE\_EVIDENCE\_REGISTRY.csv` for path, size and SHA-256 of every authority.

Supplementary Table S2. Internal secondary metrics

The complete 64-row V2 summary reports n, mean, SD, median, Q1, Q3, IQR and 10,000-resample patient bootstrap confidence intervals for eight methods and eight locked secondary metrics. Seed was 20260803. No secondary pairwise P values were added.

Supplementary Table S3. RHUH secondary metrics

All seven methods were summarized using n=39 and seed 20260810. Fixed future-blind means were Dice 0.190, soft Dice 0.151, Brier 0.0093 and AP 0.153. Canonical PCC means were 0.365, 0.238, 0.0065 and 0.309. No-smoothing means were 0.451, 0.270, 0.0058 and 0.404.

Supplementary Tables S4–S6. Development robustness

Canonical PCC mean Dice was 0.388 under clean guidance; 0.349 under Partial-50, 0.324 under Partial-25, 0.384 under FP-25, 0.368 under Shift-3 and 0.333 under Mixed guidance. Correct-target PCC exceeded shuffled-target PCC by 0.105 on average, with 39 wins and one loss. No-smoothing's clean advantage attenuated under perturbation and was not significantly worse under any tested condition in the locked robustness family.

Supplementary Table S7. Oracle-assisted controls

Target-volume-matched top-k and EIA blends use target information. Internal top-k Dice means were 0.268, 0.440 and 0.631; RHUH values were 0.207, 0.361 and 0.426. These are retrospective localization summaries.

Supplementary Table S8. Failure accounting

Development, internal confirmatory and RHUH Stage B scientific failure tables contained zero failed patients. Denominators remained 40, 113 and 39. Engineering failures before scientific access are preserved separately and were not counted as scientific failures.

Reporting checklists

The completed CLAIM 2024 matrix accompanies this supplement. TRIPOD+AI was applied only to the future-blind predictor component; PROBAST+AI informed an internal risk audit. Public URLs and final author metadata remain submission actions.

Supplementary data tables

Supplementary Table S2. Internal secondary summary

Method	Metric	n	Mean	SD	Median	Q1	Q3	CI low	CI high
Fixed	IoU_0.5	113	0.1428	0.09134	0.1363	0.08025	0.1872	0.1265	0.1601
Fixed	precision_0.5	113	0.2647	0.2096	0.2053	0.105	0.3947	0.2269	0.3036
Fixed	recall_0.5	113	0.4287	0.2542	0.3867	0.223	0.59	0.3814	0.4749

Method	Metric	n	Mean	SD	Median	Q1	Q3	CI low	CI high
Fixed	soft_Dice	113	0.1852	0.112	0.1837	0.103	0.2431	0.165	0.2062
Fixed	Brier	113	0.004523	0.004734	0.003303	0.001627	0.005121	0.003719	0.00544
Fixed	average_precision	113	0.2272	0.1624	0.188	0.09859	0.3366	0.1977	0.2581
Fixed	predicted_positive_volume	113	3.181e+04	2.191e+04	2.517e+04	1.541e+04	4.355e+04	2.784e+04	3.585e+04
Fixed	target_to_predicted_volume_ratio	113	1.25	1.837	0.5285	0.195	1.23	0.9356	1.606
Naive	IoU_0.5	113	0.1428	0.09134	0.1363	0.08025	0.1872	0.1264	0.1602
Naive	precision_0.5	113	0.2647	0.2096	0.2053	0.105	0.3947	0.2275	0.3034
Naive	recall_0.5	113	0.4287	0.2542	0.3867	0.223	0.59	0.3818	0.4759
Naive	soft_Dice	113	0.2244	0.1269	0.2243	0.1395	0.2905	0.2013	0.2482
Naive	Brier	113	0.00478	0.004989	0.003434	0.00163	0.005516	0.00392	0.005746
Naive	average_precision	113	0.2267	0.1625	0.1878	0.09874	0.3366	0.1975	0.2573
Naive	predicted_positive_volume	113	3.181e+04	2.191e+04	2.517e+04	1.541e+04	4.355e+04	2.791e+04	3.593e+04
Naive	target_to_predicted_volume_ratio	113	1.25	1.837	0.5285	0.195	1.23	0.9292	1.613
EIA-linear	IoU_0.5	113	0.1834	0.1103	0.1764	0.1088	0.2388	0.1637	0.2037
EIA-linear	precision_0.5	113	0.3132	0.2349	0.2503	0.1274	0.4548	0.2706	0.3565
EIA-linear	recall_0.5	113	0.508	0.2561	0.4872	0.3042	0.6902	0.46	0.5541
EIA-linear	soft_Dice	113	0.2602	0.1492	0.2606	0.1485	0.385	0.2329	0.2883
EIA-linear	Brier	113	0.003341	0.002776	0.002608	0.001387	0.004138	0.002857	0.003871
EIA-linear	average_precision	113	0.3611	0.2364	0.3232	0.1654	0.5269	0.3181	0.405
EIA-linear	predicted_positive_volume	113	3.223e+04	2.276e+04	2.592e+04	1.464e+04	4.6e+04	2.825e+04	3.651e+04
EIA-linear	target_to_predicted_volume_ratio	113	1.145	1.686	0.5302	0.2063	1.19	0.8602	1.471
EIA-blend-0.90	IoU_0.5	113	0.1568	0.09854	0.1464	0.09721	0.2065	0.1392	0.1756
EIA-blend-0.90	precision_0.5	113	0.294	0.224	0.2377	0.1182	0.435	0.253	0.3365
EIA-blend-0.90	recall_0.5	113	0.4342	0.2537	0.3848	0.2277	0.5965	0.3878	0.4808
EIA-blend-0.90	soft_Dice	113	0.2123	0.1212	0.2088	0.1241	0.2862	0.1901	0.2343
EIA-blend-0.90	Brier	113	0.003808	0.003966	0.002786	0.001385	0.004373	0.003123	0.004585
EIA-blend-0.90	average_precision	113	0.3588	0.1943	0.3191	0.2157	0.5073	0.3237	0.3949
EIA-blend-0.90	predicted_positive_volume	113	2.892e+04	2.039e+04	2.255e+04	1.345e+04	4.069e+04	2.528e+04	3.273e+04
EIA-blend-0.90	target_to_predicted_volume_ratio	113	1.338	1.921	0.5724	0.2242	1.389	1.012	1.709
EIA-blend-0.75	IoU_0.5	113	0.1991	0.1174	0.1869	0.1311	0.263	0.1773	0.2204
EIA-blend-0.75	precision_0.5	113	0.3793	0.257	0.3351	0.1741	0.5898	0.3327	0.4267
EIA-blend-0.75	recall_0.5	113	0.4491	0.2492	0.4115	0.235	0.6172	0.4037	0.4956
EIA-blend-0.75	soft_Dice	113	0.2532	0.1413	0.2459	0.1475	0.3632	0.2274	0.2793
EIA-blend-0.75	Brier	113	0.002876	0.002962	0.002036	0.0009949	0.003301	0.002359	0.003443
EIA-blend-0.75	average_precision	113	0.4779	0.2157	0.4498	0.3277	0.6499	0.438	0.5179
EIA-blend-0.75	predicted_positive_volume	113	2.305e+04	1.731e+04	1.924e+04	9285	3.193e+04	1.996e+04	2.637e+04
EIA-blend-0.75	target_to_predicted_volume_ratio	113	1.58	2.251	0.7209	0.295	1.957	1.195	2.017
EIA-morph	IoU_0.5	113	0.1511	0.09465	0.1358	0.09493	0.1974	0.1343	0.1688
EIA-morph	precision_0.5	113	0.2821	0.2184	0.2141	0.1117	0.4273	0.2432	0.3233
EIA-morph	recall_0.5	113	0.4297	0.2557	0.3844	0.2227	0.6041	0.3835	0.4769
EIA-morph	soft_Dice	113	0.2514	0.1372	0.2391	0.1734	0.3297	0.2259	0.2765
EIA-morph	Brier	113	0.005195	0.005021	0.003417	0.001792	0.006601	0.004324	0.006143
EIA-morph	average_precision	113	0.1059	0.08344	0.08012	0.04958	0.1501	0.09089	0.1211
EIA-morph	predicted_positive_volume	113	2.937e+04	2.184e+04	2.359e+04	1.28e+04	3.983e+04	2.544e+04	3.348e+04
EIA-morph	target_to_predicted_volume_ratio	113	3.795	26.86	0.5334	0.2071	1.328	1.024	9.061
Full PCC	IoU_0.5	113	0.3027	0.1498	0.3067	0.2104	0.3918	0.2753	0.3302
Full PCC	precision_0.5	113	0.499	0.2743	0.4785	0.2805	0.7087	0.4483	0.5492
Full PCC	recall_0.5	113	0.5404	0.215	0.5486	0.3767	0.7067	0.5006	0.5803
Full PCC	soft_Dice	113	0.2667	0.1496	0.2635	0.1626	0.3671	0.2396	0.2943
Full PCC	Brier	113	0.003102	0.004117	0.00172	0.0008422	0.003231	0.002414	0.003921
Full PCC	average_precision	113	0.4142	0.2117	0.4166	0.2552	0.5815	0.3757	0.4536
Full PCC	predicted_positive_volume	113	2.233e+04	1.85e+04	1.853e+04	8041	3.158e+04	1.904e+04	2.574e+04
Full PCC	target_to_predicted_volume_ratio	113	1.44	2.322	0.8266	0.4321	1.62	1.072	1.904
No-smoothing PCC	IoU_0.5	113	0.4958	0.2014	0.5164	0.363	0.629	0.4589	0.533
No-smoothing PCC	precision_0.5	113	0.7088	0.2561	0.7618	0.5467	0.9502	0.6607	0.754
No-smoothing PCC	recall_0.5	113	0.7034	0.244	0.7441	0.5275	0.9294	0.6575	0.7463
No-smoothing PCC	soft_Dice	113	0.3204	0.1647	0.3259	0.2144	0.4415	0.2905	0.3506
No-smoothing PCC	Brier	113	0.002583	0.004011	0.001206	0.0005939	0.002499	0.001913	0.003388
No-smoothing PCC	average_precision	113	0.6473	0.1988	0.6639	0.5326	0.8112	0.6107	0.6825
No-smoothing PCC	predicted_positive_volume	113	2.03e+04	1.91e+04	1.492e+04	6355	2.606e+04	1.7e+04	2.382e+04
No-smoothing PCC	target_to_predicted_volume_ratio	113	1.514	2.212	1.034	0.7034	1.604	1.17	1.964

Supplementary Table S3. RHUH secondary summary

Method	Metric	n	Mean	SD	Median	Q1	Q3	CI low	CI high
Fixed	Dice_0.5	39	0.1895	0.1318	0.1811	0.08101	0.2697	0.149	0.2306
Fixed	IoU_0.5	39	0.1105	0.08301	0.09956	0.04222	0.1559	0.08506	0.1365
Fixed	precision_0.5	39	0.2422	0.1968	0.2022	0.064	0.3866	0.1816	0.3029
Fixed	recall_0.5	39	0.2147	0.152	0.1606	0.1199	0.3086	0.1687	0.264
Fixed	soft_Dice	39	0.1512	0.1064	0.1456	0.06035	0.2269	0.1182	0.1838
Fixed	Brier	39	0.009284	0.004699	0.007635	0.005454	0.01239	0.007854	0.01076
Fixed	average_precision	39	0.1531	0.1213	0.1417	0.04085	0.2387	0.1162	0.191
Fixed	predicted_positive_volume	39	5.279e+04	2.36e+04	5.117e+04	3.662e+04	6.931e+04	4.571e+04	6.026e+04
Fixed	target_to_predicted_volume_ratio	39	1.494	1.55	0.933	0.6321	2.082	1.066	2.015
Naive	Dice_0.5	39	0.1895	0.1318	0.1811	0.08101	0.2697	0.149	0.2306
Naive	IoU_0.5	39	0.1105	0.08301	0.09956	0.04222	0.1559	0.08506	0.1365
Naive	precision_0.5	39	0.2422	0.1968	0.2022	0.064	0.3866	0.1816	0.3029
Naive	recall_0.5	39	0.2147	0.152	0.1606	0.1199	0.3086	0.1687	0.264
Naive	soft_Dice	39	0.179	0.1247	0.1616	0.07547	0.25	0.1405	0.2176
Naive	Brier	39	0.009386	0.005005	0.007521	0.005225	0.01299	0.00786	0.01096
Naive	average_precision	39	0.1528	0.1212	0.1396	0.04003	0.2387	0.116	0.1908
Naive	predicted_positive_volume	39	5.279e+04	2.36e+04	5.117e+04	3.662e+04	6.931e+04	4.571e+04	6.026e+04
Naive	target_to_predicted_volume_ratio	39	1.494	1.55	0.933	0.6321	2.082	1.066	2.015
EIA-linear	Dice_0.5	39	0.2461	0.153	0.2291	0.1285	0.3463	0.1983	0.2932
EIA-linear	IoU_0.5	39	0.1489	0.1014	0.1293	0.06864	0.2094	0.1175	0.1804
EIA-linear	precision_0.5	39	0.3248	0.2462	0.2727	0.1358	0.5009	0.2492	0.4024
EIA-linear	recall_0.5	39	0.2612	0.1682	0.2373	0.1646	0.3857	0.2102	0.315
EIA-linear	soft_Dice	39	0.2761	0.1357	0.2434	0.1891	0.4028	0.2337	0.317
EIA-linear	Brier	39	0.006255	0.002851	0.005547	0.003857	0.008322	0.005375	0.007154
EIA-linear	average_precision	39	0.3541	0.2166	0.2854	0.1898	0.5348	0.2868	0.4218
EIA-linear	predicted_positive_volume	39	4.744e+04	2.609e+04	4.014e+04	2.888e+04	6.252e+04	3.965e+04	5.563e+04
EIA-linear	target_to_predicted_volume_ratio	39	1.716	1.755	1.007	0.7257	2.216	1.224	2.295
EIA-blend-0.90	Dice_0.5	39	0.2059	0.1393	0.2099	0.09195	0.2979	0.1628	0.2494
EIA-blend-0.90	IoU_0.5	39	0.1215	0.08907	0.1172	0.0482	0.175	0.09425	0.1493
EIA-blend-0.90	precision_0.5	39	0.2761	0.219	0.2347	0.07691	0.4485	0.2087	0.3442
EIA-blend-0.90	recall_0.5	39	0.2211	0.1542	0.1721	0.1267	0.3218	0.1744	0.2709
EIA-blend-0.90	soft_Dice	39	0.1945	0.1145	0.1991	0.1066	0.2801	0.1593	0.2292
EIA-blend-0.90	Brier	39	0.007768	0.003957	0.006328	0.00455	0.01038	0.006563	0.009014
EIA-blend-0.90	average_precision	39	0.2722	0.1595	0.2916	0.1095	0.4156	0.2226	0.3215
EIA-blend-0.90	predicted_positive_volume	39	4.735e+04	2.248e+04	4.414e+04	3.199e+04	6.195e+04	4.065e+04	5.448e+04
EIA-blend-0.90	target_to_predicted_volume_ratio	39	1.667	1.697	1.056	0.7412	2.428	1.197	2.236
EIA-blend-0.75	Dice_0.5	39	0.2482	0.1566	0.2459	0.1243	0.3646	0.1995	0.2969
EIA-blend-0.75	IoU_0.5	39	0.1507	0.1039	0.1402	0.06627	0.2229	0.1187	0.1829
EIA-blend-0.75	precision_0.5	39	0.3631	0.2632	0.3176	0.138	0.6226	0.282	0.4447
EIA-blend-0.75	recall_0.5	39	0.2394	0.1592	0.2103	0.1392	0.3574	0.191	0.2905
EIA-blend-0.75	soft_Dice	39	0.2614	0.1309	0.238	0.1709	0.3832	0.2204	0.3007
EIA-blend-0.75	Brier	39	0.005783	0.002987	0.004744	0.003362	0.00774	0.004878	0.006726
EIA-blend-0.75	average_precision	39	0.4066	0.2094	0.4108	0.238	0.5872	0.3412	0.4701
EIA-blend-0.75	predicted_positive_volume	39	3.82e+04	2.093e+04	3.481e+04	2.26e+04	4.934e+04	3.194e+04	4.483e+04
EIA-blend-0.75	target_to_predicted_volume_ratio	39	2.095	2.031	1.305	0.9592	2.498	1.52	2.758
Full PCC	Dice_0.5	39	0.3654	0.1877	0.3913	0.2291	0.5175	0.3064	0.4214
Full PCC	IoU_0.5	39	0.2389	0.1379	0.2433	0.1295	0.3491	0.1959	0.2805
Full PCC	precision_0.5	39	0.5292	0.2845	0.5293	0.3425	0.7772	0.4403	0.6149
Full PCC	recall_0.5	39	0.3248	0.1759	0.3144	0.2341	0.4436	0.2713	0.3802
Full PCC	soft_Dice	39	0.2382	0.1445	0.2284	0.134	0.3585	0.1935	0.2817
Full PCC	Brier	39	0.006457	0.004132	0.005062	0.003083	0.008279	0.00523	0.007775
Full PCC	average_precision	39	0.3093	0.1903	0.3213	0.1503	0.4898	0.2502	0.3676
Full PCC	predicted_positive_volume	39	3.556e+04	2.366e+04	2.786e+04	1.898e+04	4.964e+04	2.839e+04	4.294e+04
Full PCC	target_to_predicted_volume_ratio	39	2.345	2.445	1.58	1.142	2.597	1.687	3.175
No-smoothing PCC	Dice_0.5	39	0.4512	0.2103	0.4658	0.3502	0.6201	0.3853	0.5142
No-smoothing PCC	IoU_0.5	39	0.3136	0.1698	0.3036	0.2125	0.4494	0.2614	0.3653
No-smoothing PCC	precision_0.5	39	0.6633	0.2889	0.7185	0.5209	0.8969	0.5707	0.748
No-smoothing PCC	recall_0.5	39	0.3898	0.2012	0.4051	0.2868	0.5306	0.3286	0.4524
No-smoothing PCC	soft_Dice	39	0.2696	0.1557	0.2777	0.1653	0.4003	0.2215	0.3165
No-smoothing PCC	Brier	39	0.005765	0.003959	0.004686	0.002675	0.007503	0.004597	0.007045
No-smoothing PCC	average_precision	39	0.4044	0.2091	0.4197	0.2725	0.5755	0.3396	0.4687

Method	Metric	n	Mean	SD	Median	Q1	Q3	CI low	CI high
No-smoothing PCC	predicted_positive_volume	39	3.36e+04	2.391e+04	2.775e+04	1.596e+04	4.568e+04	2.63e+04	4.108e+04
No-smoothing PCC	target_to_predicted_volume_ratio		2.559	2.713	1.679	1.281	2.71	1.835	3.492

Supplementary Table S4. No-smoothing robustness summary

Condition	Method	Metric	n	Mean	SD	Median	CI low	CI high
CLEAN	FULL_PCC	dice_topk	40	0.3884	0.1735	0.3775	0.3366	0.4419
CLEAN	FULL_PCC	iou_topk	40	0.2566	0.1501	0.2326	0.2133	0.3047
CLEAN	FULL_PCC	dice_fixed05	40	0.2758	0.164	0.2688	0.2279	0.3278
CLEAN	FULL_PCC	iou_fixed05	40	0.1714	0.1244	0.1553	0.1357	0.211
CLEAN	FULL_PCC	target_mass	40	1.368e+04	1.464e+04	8322	9554	1.835e+04
CLEAN	FULL_PCC	false_guidance_mass	40	0	0	0	0	0
CLEAN	FULL_PCC	outside_clean_target_mass	40	3.904e+04	2.388e+04	3.363e+04	3.197e+04	4.659e+04
CLEAN	FULL_PCC	predicted_positive_voxels	40	4.977e+04	3.051e+04	4.272e+04	4.064e+04	5.935e+04
CLEAN	NO_SMOOTHING_PCC	dice_topk	40	0.5001	0.1764	0.5134	0.4463	0.5555
CLEAN	NO_SMOOTHING_PCC	iou_topk	40	0.3529	0.1718	0.3453	0.3018	0.4077
CLEAN	NO_SMOOTHING_PCC	dice_fixed05	40	0.3263	0.1798	0.311	0.2731	0.3829
CLEAN	NO_SMOOTHING_PCC	iou_fixed05	40	0.2097	0.1428	0.1841	0.1683	0.2559
CLEAN	NO_SMOOTHING_PCC	target_mass	40	1.488e+04	1.524e+04	9762	1.058e+04	1.968e+04
CLEAN	NO_SMOOTHING_PCC	false_guidance_mass	40	0	0	0	0	0
CLEAN	NO_SMOOTHING_PCC	outside_clean_target_mass	40	3.481e+04	2.202e+04	3.027e+04	2.809e+04	4.18e+04
CLEAN	NO_SMOOTHING_PCC	predicted_positive_voxels	40	4.548e+04	2.808e+04	3.832e+04	3.699e+04	5.415e+04
FP_25	FULL_PCC	dice_topk	40	0.3842	0.1739	0.3754	0.3323	0.4386
FP_25	FULL_PCC	iou_topk	40	0.2533	0.1496	0.2311	0.2106	0.3011
FP_25	FULL_PCC	dice_fixed05	40	0.2728	0.1632	0.265	0.2251	0.3241
FP_25	FULL_PCC	iou_fixed05	40	0.1692	0.1233	0.1528	0.134	0.208
FP_25	FULL_PCC	target_mass	40	1.368e+04	1.464e+04	8322	9487	1.842e+04
FP_25	FULL_PCC	false_guidance_mass	40	240.2	206.5	201.6	178.3	305.7
FP_25	FULL_PCC	outside_clean_target_mass	40	3.972e+04	2.423e+04	3.392e+04	3.265e+04	4.724e+04
FP_25	FULL_PCC	predicted_positive_voxels	40	5.07e+04	3.094e+04	4.331e+04	4.167e+04	6.023e+04
FP_25	NO_SMOOTHING_PCC	dice_topk	40	0.4978	0.1759	0.511	0.4452	0.5527
FP_25	NO_SMOOTHING_PCC	iou_topk	40	0.3506	0.1711	0.3432	0.3008	0.406
FP_25	NO_SMOOTHING_PCC	dice_fixed05	40	0.3252	0.1792	0.3104	0.272	0.3818
FP_25	NO_SMOOTHING_PCC	iou_fixed05	40	0.2089	0.1421	0.1838	0.1678	0.2541
FP_25	NO_SMOOTHING_PCC	target_mass	40	1.488e+04	1.524e+04	9762	1.052e+04	1.972e+04
FP_25	NO_SMOOTHING_PCC	false_guidance_mass	40	489.2	391.7	382.4	375.7	612.2
FP_25	NO_SMOOTHING_PCC	outside_clean_target_mass	40	3.509e+04	2.207e+04	3.048e+04	2.851e+04	4.2e+04
FP_25	NO_SMOOTHING_PCC	predicted_positive_voxels	40	4.573e+04	2.818e+04	3.873e+04	3.718e+04	5.465e+04
MIXED	FULL_PCC	dice_topk	40	0.3328	0.1674	0.3337	0.2829	0.3849
MIXED	FULL_PCC	iou_topk	40	0.2128	0.1361	0.2003	0.1744	0.2562
MIXED	FULL_PCC	dice_fixed05	40	0.2541	0.1592	0.2333	0.2074	0.3053
MIXED	FULL_PCC	iou_fixed05	40	0.1561	0.1201	0.1321	0.1217	0.1965
MIXED	FULL_PCC	target_mass	40	1.236e+04	1.37e+04	7228	8570	1.673e+04
MIXED	FULL_PCC	false_guidance_mass	40	815	756.9	479	601.9	1058
MIXED	FULL_PCC	outside_clean_target_mass	40	3.928e+04	2.414e+04	3.384e+04	3.2e+04	4.695e+04
MIXED	FULL_PCC	predicted_positive_voxels	40	4.928e+04	3.073e+04	4.181e+04	4.032e+04	5.884e+04
MIXED	NO_SMOOTHING_PCC	dice_topk	40	0.3347	0.1703	0.3299	0.2846	0.3882
MIXED	NO_SMOOTHING_PCC	iou_topk	40	0.2146	0.1365	0.1977	0.1747	0.2583
MIXED	NO_SMOOTHING_PCC	dice_fixed05	40	0.2616	0.1646	0.2303	0.2133	0.3152
MIXED	NO_SMOOTHING_PCC	iou_fixed05	40	0.1621	0.1265	0.1303	0.1256	0.2029
MIXED	NO_SMOOTHING_PCC	target_mass	40	1.174e+04	1.314e+04	6844	8095	1.604e+04
MIXED	NO_SMOOTHING_PCC	false_guidance_mass	40	1269	1075	807.5	957.7	1615
MIXED	NO_SMOOTHING_PCC	outside_clean_target_mass	40	3.543e+04	2.214e+04	3.056e+04	2.897e+04	4.233e+04
MIXED	NO_SMOOTHING_PCC	predicted_positive_voxels	40	4.321e+04	2.824e+04	3.645e+04	3.487e+04	5.218e+04
PARTIAL_25	FULL_PCC	dice_topk	40	0.3237	0.1641	0.315	0.2765	0.376
PARTIAL_25	FULL_PCC	iou_topk	40	0.2057	0.1331	0.1869	0.1674	0.2491
PARTIAL_25	FULL_PCC	dice_fixed05	40	0.2505	0.1592	0.2199	0.2051	0.3008
PARTIAL_25	FULL_PCC	iou_fixed05	40	0.1538	0.121	0.1235	0.1195	0.1931
PARTIAL_25	FULL_PCC	target_mass	40	1.18e+04	1.327e+04	6934	8091	1.603e+04
PARTIAL_25	FULL_PCC	false_guidance_mass	40	0	0	0	0	0
PARTIAL_25	FULL_PCC	outside_clean_target_mass	40	3.822e+04	2.371e+04	3.321e+04	3.115e+04	4.571e+04
PARTIAL_25	FULL_PCC	predicted_positive_voxels	40	4.754e+04	3.012e+04	4.042e+04	3.867e+04	5.717e+04

Condition	Method	Metric	n	Mean	SD	Median	CI low	CI high
PARTIAL_25	NO_SMOOTHING_PCC	dice_topk	40	0.3322	0.1591	0.3267	0.2851	0.3822
PARTIAL_25	NO_SMOOTHING_PCC	iou_topk	40	0.211	0.1277	0.1953	0.1734	0.2526
PARTIAL_25	NO_SMOOTHING_PCC	dice_fixed05	40	0.251	0.1627	0.2192	0.2036	0.3032
PARTIAL_25	NO_SMOOTHING_PCC	iou_fixed05	40	0.1548	0.1252	0.1231	0.1188	0.1965
PARTIAL_25	NO_SMOOTHING_PCC	target_mass	40	1.099e+04	1.252e+04	6490	7522	1.508e+04
PARTIAL_25	NO_SMOOTHING_PCC	false_guidance_mass	40	0	0	0	0	0
PARTIAL_25	NO_SMOOTHING_PCC	outside_clean_target_mass	40	3.481e+04	2.202e+04	3.027e+04	2.834e+04	4.175e+04
PARTIAL_25	NO_SMOOTHING_PCC	predicted_positive_voxels	40	4.194e+04	2.799e+04	3.52e+04	3.375e+04	5.079e+04
PARTIAL_50	FULL_PCC	dice_topk	40	0.3492	0.1705	0.343	0.2976	0.402
PARTIAL_50	FULL_PCC	iou_topk	40	0.2256	0.1417	0.207	0.1838	0.271
PARTIAL_50	FULL_PCC	dice_fixed05	40	0.2604	0.1612	0.2419	0.2128	0.3112
PARTIAL_50	FULL_PCC	iou_fixed05	40	0.1606	0.1226	0.1376	0.1258	0.2015
PARTIAL_50	FULL_PCC	target_mass	40	1.255e+04	1.383e+04	7327	8604	1.704e+04
PARTIAL_50	FULL_PCC	false_guidance_mass	40	0	0	0	0	0
PARTIAL_50	FULL_PCC	outside_clean_target_mass	40	3.845e+04	2.377e+04	3.333e+04	3.142e+04	4.594e+04
PARTIAL_50	FULL_PCC	predicted_positive_voxels	40	4.836e+04	3.024e+04	4.118e+04	3.944e+04	5.761e+04
PARTIAL_50	NO_SMOOTHING_PCC	dice_topk	40	0.3886	0.1595	0.3741	0.3425	0.4395
PARTIAL_50	NO_SMOOTHING_PCC	iou_topk	40	0.2541	0.1352	0.2301	0.2152	0.2982
PARTIAL_50	NO_SMOOTHING_PCC	dice_fixed05	40	0.2758	0.1662	0.2561	0.2279	0.3298
PARTIAL_50	NO_SMOOTHING_PCC	iou_fixed05	40	0.172	0.1289	0.1469	0.135	0.2142
PARTIAL_50	NO_SMOOTHING_PCC	target_mass	40	1.224e+04	1.333e+04	7200	8536	1.668e+04
PARTIAL_50	NO_SMOOTHING_PCC	false_guidance_mass	40	0	0	0	0	0
PARTIAL_50	NO_SMOOTHING_PCC	outside_clean_target_mass	40	3.481e+04	2.202e+04	3.027e+04	2.836e+04	4.17e+04
PARTIAL_50	NO_SMOOTHING_PCC	predicted_positive_voxels	40	4.306e+04	2.792e+04	3.636e+04	3.479e+04	5.185e+04
SHIFT_3	FULL_PCC	dice_topk	40	0.3684	0.1713	0.363	0.3167	0.4213
SHIFT_3	FULL_PCC	iou_topk	40	0.2405	0.1442	0.2218	0.1989	0.2879
SHIFT_3	FULL_PCC	dice_fixed05	40	0.2699	0.1623	0.2635	0.2227	0.3203
SHIFT_3	FULL_PCC	iou_fixed05	40	0.1672	0.1228	0.1518	0.1321	0.2067
SHIFT_3	FULL_PCC	target_mass	40	1.334e+04	1.442e+04	7956	9265	1.795e+04
SHIFT_3	FULL_PCC	false_guidance_mass	40	1693	1761	962.3	1197	2268
SHIFT_3	FULL_PCC	outside_clean_target_mass	40	3.917e+04	2.39e+04	3.363e+04	3.208e+04	4.676e+04
SHIFT_3	FULL_PCC	predicted_positive_voxels	40	4.962e+04	3.057e+04	4.254e+04	4.039e+04	5.907e+04
SHIFT_3	NO_SMOOTHING_PCC	dice_topk	40	0.38	0.1725	0.3639	0.3283	0.4346
SHIFT_3	NO_SMOOTHING_PCC	iou_topk	40	0.2498	0.145	0.2225	0.207	0.2954
SHIFT_3	NO_SMOOTHING_PCC	dice_fixed05	40	0.2854	0.1691	0.2809	0.2353	0.3387
SHIFT_3	NO_SMOOTHING_PCC	iou_fixed05	40	0.1789	0.1303	0.1634	0.1411	0.2201
SHIFT_3	NO_SMOOTHING_PCC	target_mass	40	1.311e+04	1.418e+04	7843	9081	1.776e+04
SHIFT_3	NO_SMOOTHING_PCC	false_guidance_mass	40	2123	2087	1258	1525	2808
SHIFT_3	NO_SMOOTHING_PCC	outside_clean_target_mass	40	3.575e+04	2.215e+04	3.062e+04	2.926e+04	4.279e+04
SHIFT_3	NO_SMOOTHING_PCC	predicted_positive_voxels	40	4.479e+04	2.848e+04	3.759e+04	3.631e+04	5.399e+04

Supplementary Table S7a. Internal oracle-assisted summary

Method	Metric	n	Mean	SD	Median	Oracle assisted
Fixed	topk_Dice	113	0.2683	0.1577	0.2641	True
Fixed	topk_IoU	113	0.1648	0.1103	0.1522	True
Naive	topk_Dice	113	0.2686	0.1574	0.2641	True
Naive	topk_IoU	113	0.165	0.1101	0.1522	True
EIA-linear	topk_Dice	113	0.3701	0.2284	0.3387	True
EIA-linear	topk_IoU	113	0.2536	0.1929	0.2039	True
EIA-blend-0.90	topk_Dice	113	0.3388	0.1771	0.3198	True
EIA-blend-0.90	topk_IoU	113	0.2181	0.1344	0.1903	True
EIA-blend-0.75	topk_Dice	113	0.4438	0.2051	0.4128	True
EIA-blend-0.75	topk_IoU	113	0.3084	0.1797	0.2601	True
EIA-morph	topk_Dice	113	0.1907	0.1233	0.1777	True
EIA-morph	topk_IoU	113	0.1109	0.08102	0.09752	True
Full PCC	topk_Dice	113	0.4398	0.1937	0.4573	True
Full PCC	topk_IoU	113	0.3013	0.1591	0.2965	True
No-smoothing PCC	topk_Dice	113	0.6308	0.1958	0.6539	True
No-smoothing PCC	topk_IoU	113	0.4893	0.2046	0.4858	True

Supplementary Table S7b. RHUH oracle-assisted summary

Method	Metric	n	Mean	SD	Median	CI low	CI high
Fixed	topk_Dice	39	0.2066	0.1351	0.2084	0.1651	0.2484
Fixed	topk_IoU	39	0.1215	0.08608	0.1163	0.09516	0.1484
Naive	topk_Dice	39	0.2066	0.1351	0.2084	0.1651	0.2484
Naive	topk_IoU	39	0.1215	0.08608	0.1163	0.09516	0.1484
EIA-linear	topk_Dice	39	0.3435	0.2245	0.2918	0.2741	0.4134
EIA-linear	topk_IoU	39	0.2306	0.1778	0.1708	0.1764	0.2873
EIA-blend-0.90	topk_Dice	39	0.2408	0.1446	0.2619	0.1963	0.2851
EIA-blend-0.90	topk_IoU	39	0.1444	0.09449	0.1507	0.1154	0.1737
EIA-blend-0.75	topk_Dice	39	0.346	0.213	0.353	0.2799	0.4121
EIA-blend-0.75	topk_IoU	39	0.2294	0.1627	0.2143	0.1794	0.2806
Full PCC	topk_Dice	39	0.3612	0.1787	0.3843	0.3056	0.414
Full PCC	topk_IoU	39	0.2342	0.1309	0.2379	0.194	0.2734
No-smoothing PCC	topk_Dice	39	0.4263	0.1967	0.4292	0.3651	0.4844
No-smoothing PCC	topk_IoU	39	0.2896	0.1553	0.2733	0.2421	0.3364